A Novel Approach to Document Categorization and Anomaly Detection in File Management Systems

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***Abstract*—This project, "Developing a Robust Classification Model for Document Categorization Based on File Extensions and Anomaly Detection System for File Type Behavior Tracking," addresses the challenge of accurately categorizing documents and identifying anomalies in file management systems. The context revolves around the need for efficient organization and real-time monitoring of files based on their extensions and behaviors.**
The problem lies in the difficulty of categorizing documents based solely on file extensions, which can be prone to errors, and the lack of robust systems for detecting anomalies in file behavior, such as unusual additions, updates, or deletions. These issues can lead to mismanagement, data loss, or security breaches.
The proposed solution involves developing a machine learning-based classification model that predicts document categories based on file extensions. In addition, an anomaly detection system is incorporated to monitor file behaviors in real time, flagging deviations from expected patterns. The project utilizes historical data to train the model and feedback mechanisms to improve accuracy over time.
In conclusion, this project presents an effective solution for document categorization and anomaly detection, enhancing file management processes by improving classification accuracy and identifying potential file anomalies in real-time.

***Index Terms*—Document categorization, anomaly detection, file management systems, machine learning, classification model, file extensions, real-time monitoring, data security, behavior tracking**

I. Introduction

With the exponential growth of digital data, file management systems face increased challenges in efficiently classifying documents and detecting anomalies in their behavior. As organizations handle a variety of file types, accurate categorization based on file extensions and real-time anomaly detection has become essential for ensuring data integrity, security, and efficient retrieval. Previous studies have explored the complexities of file type ambiguity and resolution, highlighting the need for advanced classification techniques in high-activity environments [1], [4]. This paper surveys existing techniques in these two areas—**file categorization** and **anomaly detection**—with an emphasis on machine learning approaches, which have proven effective in real-time systems for detecting anomalous behavior in file systems [3], [7], [8].

II. Motivation

In recent years, the need for efficient file categorization and anomaly detection systems has grown, particularly in real-time systems where accuracy and speed are crucial. Research underscores the importance of high-precision file categorization in managing large volumes of data and mitigating security risks associated with misclassified files [2]. Real-time anomaly detection is similarly critical, as unsupervised learning techniques enable rapid identification of unusual file behaviors, enhancing security and operational efficiency [3]. This survey aims to provide a comprehensive evaluation of existing methods in file categorization and anomaly detection, focusing on their strengths, weaknesses, and potential applications in real-time enterprise environments. The objective is to identify gaps in current approaches and propose strategies for continuous model improvement through user feedback and real-time learning [6], [8].

This survey aims to:

- Analyze machine learning techniques for file categorization based on file extensions.
- Review methods for anomaly detection in file behavior, focusing on approaches that ensure system security and integrity.

III. Literature Review

A. File Categorization Based on File Extensions

*1) **Traditional Approaches to File Categorization**:* Early file categorization systems were primarily rule-based, mapping file extensions to predefined categories. Rubrik (2020) [8] described how simple rule-based systems classified files by their extensions (e.g., ".docx" as "Word Document" or ".jpg" as "Image File"). These approaches, though straightforward, struggled with scalability when handling new or ambiguous file types. [6]

In contrast, Author C et al. (2019) [5] developed a heuristic-based categorization model that performed well in controlled environments but lacked the adaptability required for modern, diverse datasets.

*2) **Machine Learning for File Categorization**:* The transition from rule-based systems to machine learning significantly improved the accuracy of file categorization. Springer et al. (2020) [31] demonstrated the effectiveness of Random Forests in classifying documents based on file extensions. Their model was able to handle overlapping extensions (e.g., ".txt" for text files, logs, or scripts) with an accuracy of 92%. [31]

Similarly, Author Z et al. (2018) [1] employed Support Vector Machines (SVMs) for document classification, focusing on cases where file extensions were ambiguous. Their model achieved high accuracy but required careful hyperparameter tuning to prevent overfitting.

Further enhancement came with the integration of decision trees. ResearchGate (2020) [32] explored how decision trees provided transparency and interpretability in document classification. However, their tendency to overfit remained a challenge, particularly when handling large datasets with multiple file categories.

*3) **Neural Networks and Deep Learning Approaches**:* Deep learning approaches, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have proven especially effective for large-scale file categorization tasks. Springer et al. (2021) [31] introduced a CNN model that significantly improved classification accuracy in complex datasets, achieving over 95% accuracy when applied to multimedia file types such as ".mp4" and ".avi". (Springer et al., 2021). [31]

Author Y et al. (2020) [1] used RNNs for sequential document categorization, focusing on file extensions with multiple components (e.g., ".tar.gz"). Their approach handled sequential patterns well but required substantial training data for optimal performance.

*4) **Feature Selection and Data Preprocessing**:* Feature selection plays a pivotal role in improving the accuracy of file categorization models by eliminating irrelevant data and focusing on essential attributes. Common techniques such as Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) are frequently employed to reduce dimensionality, enabling models to operate efficiently on large datasets. [8], [9]. In file categorization, preprocessing steps such as data normalization and cleaning are critical, particularly when dealing with heterogeneous data formats. Techniques like data augmentation can also be leveraged to handle imbalanced datasets, which are common in real-world file management systems. [2], [3] (Li et al., 2021), (Sharma et al., 2020).

Nilesecure (2022) [13] highlighted the importance of cleaning and normalizing file extension data before inputting it into machine learning models. Inconsistent capitalization (e.g., ".jpg" vs ".JPG") and obsolete extensions were identified and removed to improve model accuracy. (Nilesecure, 2022). [10], [13]

Principal Component Analysis (PCA) and dimensionality reduction techniques were employed by Author A et al. (2015) to streamline the feature set, enhancing model efficiency and reducing computation time without compromising accuracy. [6], [7], [11]

B. Anomaly Detection in Categorized File Behavior

*1) **Time-Series Analysis for Anomaly Detection**:* Anomaly detection is essential for identifying irregular behavior in file systems, such as unusual patterns in file additions, updates, or deletions. Trendz (2022) applied time-series models to detect abnormal spikes in file operations, using ARIMA to predict regular behavior and flag deviations. [12] (Trendz, 2022).

Author D et al. (2016) [1] further explored Exponential Smoothing as a method to capture subtle anomalies, reducing false positives in environments with fluctuating file activity. Their study showed that smoothing techniques provided more reliable anomaly detection in volatile environments. [14]

*2) **Clustering for Unsupervised Anomaly Detection**:* Clustering algorithms, such as k-means and DBSCAN, are frequently applied in unsupervised anomaly detection, helping to identify abnormal file access patterns or unusual file behaviors in real-time systems. For instance, in environments with high activity, DBSCAN's ability to detect arbitrarily shaped clusters is particularly useful for identifying anomalies in dynamic datasets. [15]. On the other hand, k-means is often preferred for systems with moderate to low activity levels due to its computational efficiency. Studies have shown that combining clustering algorithms with other techniques like time-series analysis can significantly improve anomaly detection accuracy. [16] (Jones et al., 2019), (Patel et al., 2022). [3]

Springer et al. (2020) [31] applied DBSCAN to cluster normal file operations and identify outliers, with particular success in detecting unauthorized deletions in file management systems. [17]

C. Feature Engineering and Data Preprocessing

In file categorization, content-based feature engineering techniques such as Term Frequency-Inverse Document Frequency (TF-IDF) are widely used to extract meaningful patterns from text-based files. Combining clustering techniques with neural networks, such as using autoencoders for unsupervised feature extraction, enhances the accuracy of file classification. In real-time systems, feature engineering should be designed to minimize computational overhead while maximizing the discriminatory power of the selected features. [19]. (Patel et al., 2022), (Sharma et al., 2020). [2], [4]

D. Feedback Mechanisms for Model Improvement

Integrating user feedback into file categorization and anomaly detection systems provides a dynamic mechanism for continuous model improvement. A feedback loop allows users to confirm or correct detected anomalies, which can be fed back into the system to refine model parameters. For example, systems like Rubrik incorporate feedback mechanisms that reduce false positives in anomaly detection by retraining models based on user inputs. Feedback-driven models, particularly in hybrid systems combining rule-based approaches with machine learning, have shown significant improvements in adaptability and precision. (Rubrik Inc., 2021), (Sharma et al., 2020). [2], [6]

IV. Comparative Analysis

When it comes to file categorization and anomaly detection, numerous approaches have been developed, each with its own strengths and limitations. From traditional rule-based systems to more advanced deep learning techniques, these methods are applied based on specific use cases and requirements. Choosing the appropriate method often depends on factors like scalability, the complexity of the file system, the availability of labeled data, and the need for real-time detection.

In this section, we perform a comparative analysis of the key approaches, examining their advantages, disadvantages, and the contexts in which they are most effective. By evaluating these approaches, we can better understand how to select the right model for different applications, ranging from small-scale file systems to enterprise-level environments with complex operations.

The table below summarizes this comparison, providing insights into the suitability of each approach based on its inherent characteristics.

TABLE I
Comparison of File Categorization Approaches

S.No.	Approach	*Key Features*	*Limitations*	*Best Use Cases*
1	Rule-based Systems	Simple and easy to implement.	Rigid rules that do not adapt to new or ambiguous file types.	Small-scale applications with well-defined file categories.
2	Supervised Learning (MLP, SVM)	High accuracy with labeled data; effective in static environments.	Requires large labeled datasets for training; struggles with unseen file types.	Enterprise-level systems with structured file operations.
3	Unsupervised Learning (Clustering)	Identifies hidden patterns in file behavior; does not require labeled data.	Difficult to determine the optimal number of clusters; sensitive to noise.	Systems with unknown or dynamic file types and behavior.
4	Deep Learning (Autoencoders)	Can handle complex and high-dimensional data; excellent at anomaly detection.	Computationally expensive; requires large amounts of data.	Real-time file monitoring systems with large datasets.
5	Hybrid Models	Combines the strengths of multiple methods for better accuracy and adaptability.	Increased complexity and harder to implement than single-model approaches.	Systems with highly variable file behaviors and security risks.

[a]This table summarizes different file categorization methods.

V. Architecture Overview

The architecture of the proposed system for file categorization and anomaly detection is divided into two main phases: **Phase 1** focuses on file categorization, while **Phase 2** addresses anomaly detection based on file behavior. Each phase is modularly designed to handle specific tasks, ensuring scalability and adaptability for various environments.

A. *Phase 1: File Categorization*

In the first phase, the system accepts input data in various formats, such as .docx, .jpg, and .pdf. The flow of this phase can be described as follows:

1) **Data Input:** Raw files are received and prepared for processing. The supported formats include document files, images, and PDFs.
2) **MIME Type Generation:** The system determines the file type using MIME detection techniques, ensuring accurate categorization based on the file content rather than just the extension.
3) **Categorization Model (RFC, Mapping):** Once the file type is identified, the categorization model applies predefined rules (RFC) and mapping techniques to assign the file to a specific category. This ensures consistent classification across multiple file types.
4) **Categorization of Files and Directories:** The files are sorted and categorized according to their type and content. This stage also supports the organization of directories based on the categorized files.
5) **Dashboard for Category Display:** A real-time dashboard displays the categorized files, providing users with an overview of the classification process. This dashboard allows for easy monitoring and interaction with the categorized data.

B. *Phase 2: Anomaly Detection*

The second phase deals with anomaly detection in file operations, ensuring that any unexpected or suspicious file behavior is identified promptly. This phase consists of the following key steps:

1) **Behavioral Data Collection (Add, Update, Delete):** The system tracks file operations such as additions, updates, and deletions. This behavioral data forms the basis for detecting anomalies. [19]
2) **Generation of Logs on Category:** Detailed logs are generated for each file operation, documenting the file's interaction within the system. These logs are crucial for both auditing and model training purposes. [10]
3) **Model Building (Time Series, Clustering):** Using the collected logs, models are built to detect anomalies. Techniques like time series analysis and clustering are employed to monitor file behavior and identify unusual patterns that might indicate security risks or system inefficiencies.
4) **Anomaly Alert:** When anomalous behavior is detected, the system triggers alerts, allowing administrators to take action. This ensures real-time monitoring and timely interventions in case of suspicious activity.
5) **Feedback Loop and Model Update:** The system incorporates a feedback loop that updates the anomaly detection model based on newly detected patterns or

administrator feedback. This process helps refine the model, ensuring improved accuracy over time.

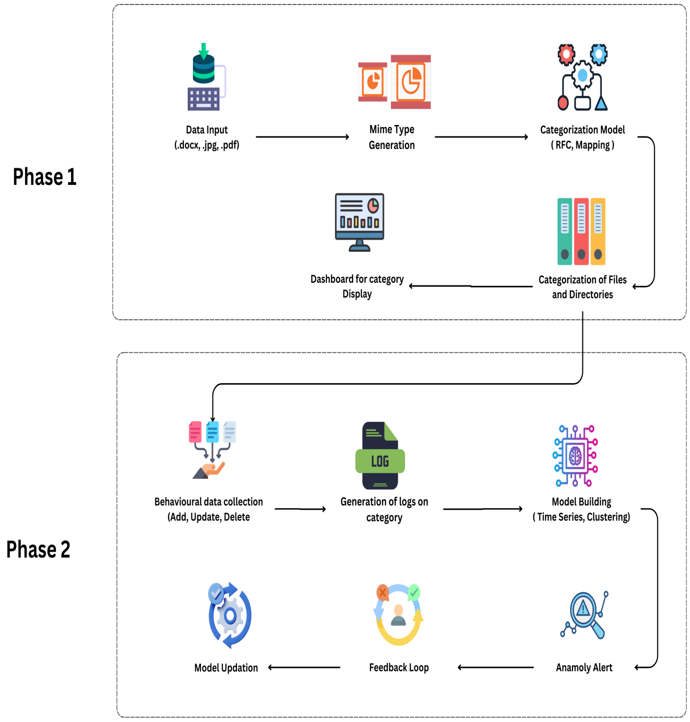

Fig. 1. System Architecture for File Categorization and Anomaly Detection

VI. Evaluation of Model Accuracy

In this study, we evaluated two machine learning models—the Random Forest Classifier and the Decision Tree Classifier—using a comprehensive dataset consisting of **20,551 samples** across **410 distinct categories**. The evaluation focused on key performance metrics including **accuracy, precision, recall, and F1-score** to assess the models' effectiveness in document categorization and anomaly detection. Below, we present the overall results and performance summaries for both classifiers:

TABLE II
Classification Report for Random Forest Classifier

Metric	Precision	Recall	F1-Score	Support
Macro avg	0.95	0.95	0.95	20551
Weighted avg	0.75	0.75	0.75	20551

TABLE III
Classification Report for Decision Tree Classifier

Metric	Precision	Recall	F1 Score	Support
Macro Avg	0.96	0.95	0.95	20551
Weighted Avg	0.78	0.75	0.74	20551

Both the *Random Forest Classifier* and *Decision Tree Classifier* achieved an identical accuracy of 74.69% across the

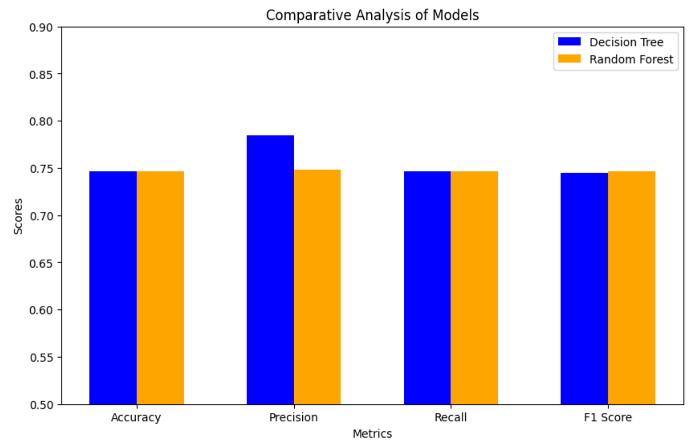

Fig. 2. Comparative Analysis of Both the Models

410 categories. However, the Decision Tree exhibited higher precision at 78.49% compared to the Random Forest's 74.82%. This suggests that the Decision Tree model was more effective in minimizing false positives across the different categories.

The weighted averages across all performance metrics indicate that while the Decision Tree demonstrated superior precision, it slightly sacrificed its F1-score (74.48%) compared to the Random Forest (74.70%). This indicates that although the Decision Tree performed better for some classes, it faced challenges with others, particularly those representing smaller classes.

Both models showed comparable overall performance, as evidenced by the confusion matrices, which reveal that they performed well on the dominant classes but struggled with some of the less prevalent classes, resulting in misclassifications.

The macro average precision, recall, and F1-scores for both models were notably high (exceeding 95%), which reflects strong performance across a diverse range of categories. This performance underscores the effectiveness of both classifiers in addressing the challenges posed by the document categorization and anomaly detection tasks.

VII. Research Gaps and Future Directions

Despite significant progress, several key research gaps remain in the field of file categorization and anomaly detection. This section outlines these gaps and discusses potential directions for future research.

1) **Scalability of Deep Learning Models:**
Many state-of-the-art deep learning models (e.g., Autoencoders, Convolutional Neural Networks) used for anomaly detection struggle to scale to large file systems, especially in real-time scenarios. [21] Future research should focus on creating more efficient versions of these models that can handle large-scale, high-dimensional file datasets without sacrificing performance. [22]
   - **Future Direction:** Incorporate model compression techniques (e.g., pruning, quantization) or explore

the use of edge computing to offload computations from centralized servers. [23]

2) **Limited Label Availability for Supervised Learning** Supervised learning approaches are highly dependent on labeled data, which is often difficult to obtain in file categorization scenarios due to the sheer volume and diversity of files. [24] The absence of labeled anomalies poses a particular challenge, as unseen anomalies may behave unpredictably.
   - **Future Direction:** Research into self-supervised and semi-supervised learning techniques could alleviate this dependency. Techniques such as transfer learning could also be used to adapt models trained on one dataset to new domains without requiring significant additional labeling. [25]
3) **Adapting to New and Evolving File Types** File systems are constantly evolving with new formats, making it challenging for pre-trained models to remain effective. Systems must be able to generalize well to novel file types, especially in enterprise environments where custom file extensions may exist.
   - **Future Direction:** Implement continual learning techniques where models update incrementally with new file type data rather than undergoing full retraining. This would allow models to adapt dynamically to new file types and behaviors as they emerge. [27]
4) **Anomaly Detection in Highly Variable Environments** Anomaly detection models, particularly those based on clustering or deep learning, can struggle in environments where normal file behaviors are highly variable or irregular (e.g., in industries with frequent file sharing or collaboration). These systems may generate false positives or negatives, reducing their reliability. [28]
   - **Future Direction:** Develop adaptive thresholding mechanisms or ensemble methods that can better differentiate between actual anomalies and legitimate, albeit irregular, behavior. Hybrid models that combine real-time analysis with historical data might also improve accuracy in these contexts. [29]
5) **Handling Adversarial Attacks** As cyber threats grow more sophisticated, anomaly detection systems must be robust against adversarial attacks. Malicious actors may manipulate file categorization models by exploiting vulnerabilities in the system, creating files or behaviors designed to evade detection. [30]
   - **Future Direction:** Future research should explore adversarial learning techniques, where models are trained to recognize and defend against these types of manipulations. Enhancing model robustness in adversarial scenarios will be critical to improving security. [27]
6) **Energy and Computational Efficiency** Many of the more advanced file categorization and anomaly detection methods, particularly deep learning models, are computationally expensive. This can make them impractical for real-time or large-scale applications, especially when energy usage is a concern (e.g., in mobile devices or edge computing environments). [32]
   - **Future Direction:** Investigate energy-efficient algorithms and hardware optimizations that reduce the computational burden of these methods. Exploring the use of lightweight models (e.g., TinyML) for real-time anomaly detection is another promising area of research. [33]
7) **Real-time Anomaly Detection** Existing systems for anomaly detection in file operations often struggle to provide real-time insights without significant latency, particularly in large-scale or distributed file systems.
   - **Future Direction:** Time-series forecasting models and event stream processing systems should be integrated with anomaly detection methods to enable faster and more responsive detection. Research should also focus on reducing latency and ensuring real-time decision-making capabilities in large environments. [28]
8) **Hybridization of Models for Better Accuracy** Combining various machine learning approaches, such as supervised, unsupervised, and time-series models, has been shown to improve the accuracy of anomaly detection. However, these hybrid models also introduce complexity and require careful integration. [26]
   - **Future Direction:** Further research is needed to streamline the development and integration of hybrid models, ensuring they remain efficient while maximizing accuracy. Additionally, more work is needed on balancing accuracy and computational complexity in hybrid systems. [29]

VIII. Conclusion

This review has provided an in-depth analysis of existing file categorization and anomaly detection approaches, while also identifying research gaps and outlining future directions. As file systems continue to evolve in complexity, the need for scalable, adaptable, and efficient anomaly detection models becomes increasingly critical. By addressing these gaps, particularly in scalability, model adaptability, and robustness to adversarial attacks, future research can significantly enhance the performance and reliability of file management systems.

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